The Optimizing Universe:
Physics as Informational Network Optimization

Kuro yamada

Abstract

Modern physics relies on hardcoded parameters such as "mass" and the "gravitational constant G," sustaining profound contradictions between macroscopic general relativity and microscopic quantum mechanics. This paper proposes a radically new informational-geometric paradigm that redefines the universe as a "fully self-learning distributed network" devoid of any external programmer or initialization (boot) process.

We define space not as a mere background, but as an information network possessing physical density ρ. This spatial density determines the network's clock frequency (information update capacity $c(\rho)$), meaning regions with greater mass experience processing delays. The network operates autonomously to continuously monitor and minimize ($\delta T = 0$) the "total informational delay cost T," which is the sum of communication latency (static delay) and the synchronization cost associated with routing modifications (dynamic delay).

The three primary conclusions proven in this paper are:

1. **Gravity as Gradient Descent:** Universal gravitation is not an attractive force between masses. Rather, it is the macroscopic manifestation of the weight-updating protocol $\nabla(1/c) = -\nabla\Gamma$. When a gradient in communication latency ($1/c(\rho)$) occurs, the network updates the geometric weight (Γ) in real-time to offset the error (delay cost). Planetary orbits are simply optimization routes where the loss function has reached a local minimum [1].

2. **Quantum Mechanics as Exception Handling and Resolution Limits:** Wave function collapse is not an occult phenomenon driven by an observer's consciousness, but rather a "forced synchronization timeout error (exception handling)" triggered when the network's synchronization computational cost exceeds the local allowable time (timeout limit τ). Furthermore, Heisenberg's uncertainty principle represents a pure information-processing trade-off—a sampling resolution limit (aliasing) of the information network [2, 3].

3. **Elimination of Dark Matter and Dark Energy (Bug Fix):** The greatest mysteries of cosmology [4, 5] are neither unknown particles [6, 7] nor exotic energies. They are "macroscopic artifacts (illusions)" arising because classical physics algorithms failed to account for the dynamic synchronization cost (delay density Λ) incurred when information traverses spatial inhomogeneities.

In conclusion, the universe possesses neither an absolute "correct answer" nor a "power button" (Big Bang) to initiate it. It is an eternal feedback loop (state machine) relentlessly executing a communication cost minimization algorithm. By taking the "subtractive" approach to physical laws to its absolute limit, the SPI/DUP/GU framework reveals that the fundamental architecture of the cosmos is structurally identical to modern AI and machine learning optimization algorithms.

1 The Hardware Architecture of the Universe

Classical physics treats "space" as an empty void and "mass" and "time" as absolute, independent parameters. However, when viewing the universe as a self-learning distributed network, these concepts translate directly into the **hardware constraints of an information processing system**.

Based on the SPI/DUP theoretical framework, this section demonstrates that the physical laws of the universe are implemented using the exact same structural constraints as computer architecture: clock frequency, memory allocation, and communication latency.

1.1 Spatial Density ρ as Clock Frequency

Einstein's theory of relativity predicts that "time slows down in strong gravitational fields." From a systems engineering perspective, however, "time" itself is not stretching. Rather, the local network is experiencing a **drop in clock frequency (processing delay)**.

In this framework, space is not a mere background, but an informational substrate possessing physical density ρ. This spatial density determines the network's baseline clock frequency, or its ability to update states, denoted as $c(\rho)$.

- **High-Density Regions (e.g., near a star):** Information is densely packed, reducing the network's update capacity $c(\rho)$. This is analogous to a server processing a heavy load, causing its clock frequency to drop and inducing processing delays (perceived macroscopically as gravitational time dilation).
- **Low-Density Regions (e.g., deep space):** The computational load is light, allowing the clock frequency to rise and information to update rapidly.

Thus, cosmic time is not a continuous, flowing river, but merely the execution cycle of the spatial network's informational updates.

1.2 Mass m as Synchronization Memory (Informational Resistance)

The concept of "mass," long held sacred in physics, reveals an entirely different nature when viewed through the lens of computer science. Mass is not the intrinsic "weight" of a substance; it is merely the **data size (memory allocation)** required to synchronize a state across the network.

The Dual Update Principle (DUP) redefines mass using the following simple equation:

$$m = \rho u \quad \text{(Mass = Spatial Density} \times \text{Informational Update Time)} \tag{1}$$

This equation indicates that mass is the "informational resistance" encountered when redistributing data within the system.

- **The Reality of Inertia:** When velocity information is input into a multi-atomic object, a time lag occurs before this data is synchronized across the entire network (the object).
- **Massive Objects:** These require a larger data size (memory) to be synchronized, resulting in a longer propagation delay for the information update. This "synchronization delay" is precisely what we experience macroscopically as "inertia" or "weight".

Pushing a heavy object is difficult not because the Earth is pulling it, but because *broadcasting a massive data block across the local network requires immense computational resources and time.*

1.3 The Speed of Light $c(\rho)$ as Communication Latency (Ping)

The "constancy of the speed of light," a fundamental premise of modern physics, is an unnatural constraint from a networking perspective. In this theory, the speed of light $c(\rho)$ is not an absolute cosmic speed limit, but rather the **baseline communication latency (Ping)** of the network, which depends heavily on the spatial density ρ.

Light propagation is the process of routing information packets across the spatial network. A finite speed of light implies that the universe possesses a finite network bandwidth. When light passes near a massive object (a high-density region with poor Ping), communication latency increases. As detailed in the next chapter, light bends near stars not because "spacetime is curved," but because the universe's routing algorithm autonomously selects the path of least processing time, deliberately bypassing heavy, high-latency nodes (high-density space) to minimize total delay.

2 Gravity as Gradient Descent: The Universal Loss Function

In artificial intelligence and machine learning, models "learn" by defining a loss function (error or cost) and continuously updating network weights to minimize it through gradient descent. Remarkably, the physical laws governing the universe operate on this exact same optimization algorithm.

Newtonian "attractive force" is an illusion. What we observe as gravity is merely the macroscopic manifestation of a localized routing optimization (gradient descent) executed by the spatial network to minimize communication delays.

2.1 The Universal Loss Function: Total Informational Latency T

The spatial network continuously monitors and optimizes a specific loss function: the total informational delay cost T incurred when data (matter or light) propagates through space. Based on the GU (Geometric Universe) framework, this loss function is defined by the following integral:

$$T[\gamma] = \int_\gamma \frac{ds}{c(\rho(s))} + \int_\gamma \Gamma(s)ds \tag{2}$$

From a computer science perspective, this equation is structurally identical to a machine learning loss function featuring a primary cost and a regularization penalty.

- **Static Cost (Communication Latency):** The first term represents the baseline data transfer cost depending on spatial density ρ. In high-density regions, the clock frequency drops, increasing latency. The universe's routing algorithm naturally avoids these high-latency nodes. This autonomous "latency avoidance" is the true mechanism behind the refraction of light near stars [6, 7].
- **Dynamic Cost (Regularization Penalty):** The second term, Γ, is the synchronization cost incurred when a trajectory changes direction. If a path bends too sharply to avoid latency, the network must expend additional computational resources to synchronize the new directional state. This acts precisely like an L1/L2 regularization penalty, suppressing excessive curvature.

2.2 The Gradient Law as Weight Updating (Backpropagation)

To minimize the loss function T (where $\delta T = 0$), the universe must update its network parameters. The protocol for this weight update is the fundamental master equation of the theory, the Gradient Law:

$$\nabla \left(\frac{1}{c(\rho)} \right) = -\nabla\Gamma \tag{3}$$

This is not an equation of physical force; it is an algorithm for error backpropagation. The left side represents the spatial gradient (error) in communication latency. The right side dictates that the network must autonomously update its geometric weight (Γ) in real-time, executing a gradient descent ($-\nabla\Gamma$) to perfectly offset the latency error.

2.3 Orbital Motion as Local Minima

This optimization perspective radically redefines celestial mechanics. A planet does not orbit a star because it is tethered by an invisible gravitational pull.

In the GU framework, there is a strict 1:1 symmetric partition between the static latency gradient and the dynamic signal-propagation delay. An orbit is simply the equilibrium state where the static input from the density gradient perfectly balances the dynamic update cost of changing direction [8, 9, 10].

In optimization terms, a planetary orbit is a **local minimum**—the exact path where the network has fully executed its gradient descent, the loss function T is minimized, and no further weight updates are required.

3 Exception Handling and Network Limits: The Quantum Regime

In distributed computing, perfect real-time synchronization across all nodes with finite bandwidth is physically impossible. The spatial information network of the universe is no exception.

In the GU framework, the deeply counterintuitive phenomena of quantum mechanics—such as the uncertainty principle and wave function collapse—are completely demystified. They are not occult phenomena driven by observer consciousness. Rather, quantum mechanics is simply the study of the spatial network's **processing limits (sampling theorems) and its built-in error-handling protocols (exception handling)**.

3.1 The Uncertainty Principle as a Sampling Limit (Aliasing)

Heisenberg's uncertainty principle is traditionally taught as a fundamental mystery. From an information engineering perspective, however, it is merely a network sampling limit.

The spatial network possesses a minimum resolution scale. Momentum represents the update frequency of a localized particle.

- **Bandwidth Trade-off:** Attempting to process high-frequency momentum updates requires rapid, large-scale synchronization across the mesh. This massive data traffic consumes local computational resources, forcibly degrading the precision of the local spatial resolution.

- **Geometric Aliasing:** The uncertainty relation is therefore an emergent informational constraint [2, 3, 11]. It reflects the impossibility of achieving infinite resolution in both the spatial and frequency domains simultaneously across a discrete informational substrate.

3.2 Wave Function Collapse as a Synchronization Timeout

Why does a superposition "collapse" into a single state? The classical interpretation invokes the philosophical "consciousness of the observer." The GU framework replaces this with a rigorous computational limit: **a synchronization timeout**.

Maintaining multiple asynchronous states requires continuous computational overhead. When a massive object attempts to synchronize two contradictory geometric states separated by a distance l, the required computational time can exceed the network's local allowable timeout limit τ. This local timeout threshold is strictly defined in the GU framework as:

$$\tau = \frac{2\hbar l}{mR_s c^2} \tag{4}$$

- **System Crash Prevention:** When the required synchronization time surpasses τ, the network experiences a critical timeout error.
- **Exception Handling:** To prevent a systemic crash, the network's fail-safe protocol triggers an "Exception Handling" routine. It forcibly terminates the asynchronous threads, randomly selecting and overwriting the system with one definitive geometric state.

4 Garbage Collection and the Eternal Loop: The Steady-State Architecture

In computer science, if allocated memory is never freed, the system inevitably succumbs to resource exhaustion (a memory leak) and crashes—a fate analogous to the thermodynamic "heat death" of the universe [12]. However, the spatial information network implements a flawless, continuous Garbage Collection (GC) protocol to reclaim unused resources.

4.1 Black Holes as Garbage Collectors (GC)

In general relativity, a black hole is an infinite curvature singularity that destroys matter. From a systems engineering perspective, it is a crucial network maintenance routine [3]. When complex data structures (atoms) enter the extreme high-density environment of a black hole, the computational tension required to synchronize their internal state diverges.

- **Data Disintegration:** Unable to maintain synchronization, the complex atomic data structures collapse into their most primitive informational state: hydrogen protons and freed spatial units.
- **Memory Deallocation:** The space that was heavily allocated to maintain the heavy atomic structures is released back into the cosmic network as free spatial memory.

4.2 The Absence of a Boot Sequence (The Illusion of the Big Bang)

A computer requires an initialization or "boot" sequence only when transitioning from a powered-down state to an active state. However, the universe is a **steady-state running process** perpetually executing the optimization protocol $\nabla(1/c) = -\nabla\Gamma$.

- **No Big Bang:** The universe was never "turned on." It is an eternal state machine that continuously recomputes its present configuration.
- **Redshift as Network Latency:** The observed cosmological redshift, traditionally interpreted as the physical expansion of the universe [5, 13], is merely a "Spatial-Density Doppler Effect." It is an optical illusion generated by the accumulated informational latency as light signals traverse varying density gradients across the cosmic network [4, 14, 15, 16]. We are observing signals delayed by the heavy routing topology of a static network.

5 Conclusion: The Self-Learning Universe

This paper establishes that the universe requires no external programmer, no absolute time, and no hardcoded magical constants. By adopting the SPI/DUP/GU framework, we successfully map the fundamental laws of physics directly onto the architecture of a self-learning distributed network:

- **Gravity** is the gradient descent (weight update) executed by the network to minimize communication latency.
- **Mass** is the synchronization memory (data size) resisting rapid state updates.
- **Quantum Mechanics** is the strict protocol for network sampling limits and synchronization timeout errors (exception handling).
- **Black Holes** are the garbage collectors ensuring eternal memory reallocation.

Newton's "attractive force" and Einstein's "curved spacetime" were merely macroscopic illusions—approximations of an underlying routing algorithm minimizing total delay cost. The ultimate "Theory of Everything" is not found in an ever-expanding zoo of hypothetical subatomic particles. It is found in the supreme elegance of subtractive physics: the universe is a perpetual, self-referential algorithm executing the simplest possible cost-minimization code on the substrate of space.

References

[1] H. Goldstein. *Classical Mechanics*. Addison-Wesley, 3rd edition, 2001.

[2] G. 't Hooft. Dimensional reduction in quantum gravity, 1993.

[3] L. Susskind. The world as a hologram. *Journal of Mathematical Physics*, 36(11):6377–6396, 1995.

[4] S. Alam et al. The eleventh and twelfth data releases of the sloan digital sky survey. *Astrophysical Journal Supplement Series*, 219(1):12, 2015.

[5] A.G. Riess et al. Large magellanic cloud cepheid standards provide a 1% foundation for the determination of the hubble constant. *The Astrophysical Journal*, 876(1):85, 2019.

[6] J.-P. Kneib, Y. Mellier, B. Fort, and G. Mathez. The distribution of dark matter in the cluster of galaxies abell 370. *Astronomy and Astrophysics*, 273:367–376, 1993.

[7] A. Zitrin, A. Fabris, J. Merten, D. Coe, E. Katz, M. Postman, C. Grillo, A. Mercurio, M.C. Carrasco, S. Cantalupo, L.A. Moustakas, T. Anguita, B. Frye, W. Zheng, T. Broadhurst, M. Meneghetti, R. Bouwens, D. Bradley, H. Ford, D. Lemze, J. Richard, A.G. Riess, S.A. Rodney, P. Rosati, S. Seitz, and A. Wel. The hubble frontier fields: A comparison of lens models for abell 2744. *The Astrophysical Journal*, 811(1):47, 2015.

[8] R.B. Tully and J.R. Fisher. A new method of determining distances to galaxies. *Astronomy and Astrophysics*, 54:661–673, 1977.

[9] S.S. McGaugh. The baryonic tully-fisher relation of gas-rich galaxies as a test of λcdm and mond. *The Astronomical Journal*, 143(2):40, 2012.

[10] F. Lelli, S.S. McGaugh, and J.M. Schombert. Sparc: Mass models for 175 disk galaxies with spitzer photometry and accurate rotation curves. *The Astronomical Journal*, 152(6):157, 2016.

[11] R. Bousso. A covariant entropy conjecture. *Journal of High Energy Physics*, 1999(07):004, 1999.

[12] P.J.E. Peebles. *The Large-Scale Structure of the Universe*. Princeton University Press, 1980.

[13] D.W. Hogg. Distance measures in cosmology, 1999.

[14] M. Tegmark, A. Oliveira-Costa, and A. Hamilton. A high resolution foreground cleaned cmb map from wmap. *Physical Review D*, 68:123523, 2003.

[15] K. Land and J. Magueijo. Examination of evidence for a preferred axis in the cosmic radiation anisotropy. *Physical Review Letters*, 95:071301, 2005.

[16] Planck Collaboration. Planck 2018 results. vii. isotropy and statistics of the cmb. *Astronomy & Astrophysics*, 641:7, 2020.